Series

What could an infinite series mean?

$$\sum_{n=1}^{\infty} x_n = x_1 + x_2 + x_3 + \cdots$$

There is a potential philosophical problem here.

Def The ~~partial~~ k^{th} partial sum of an infinite series $\sum_{n=1}^{\infty} x_n$ is the finite sum

$$\sum_{n=1}^{k} x_n = x_1 + x_2 + \cdots + x_k.$$

Def 2.21

$$\sum_{n=1}^{\infty} x_n := \lim_{k \to \infty} \left(\sum_{n=1}^{k} x_n \right)$$

Series is convergent if this limit exists

Clearly

$$\sum_{m=1}^{\infty} m = 1+2+3+4+\dots$$

this is divergent.

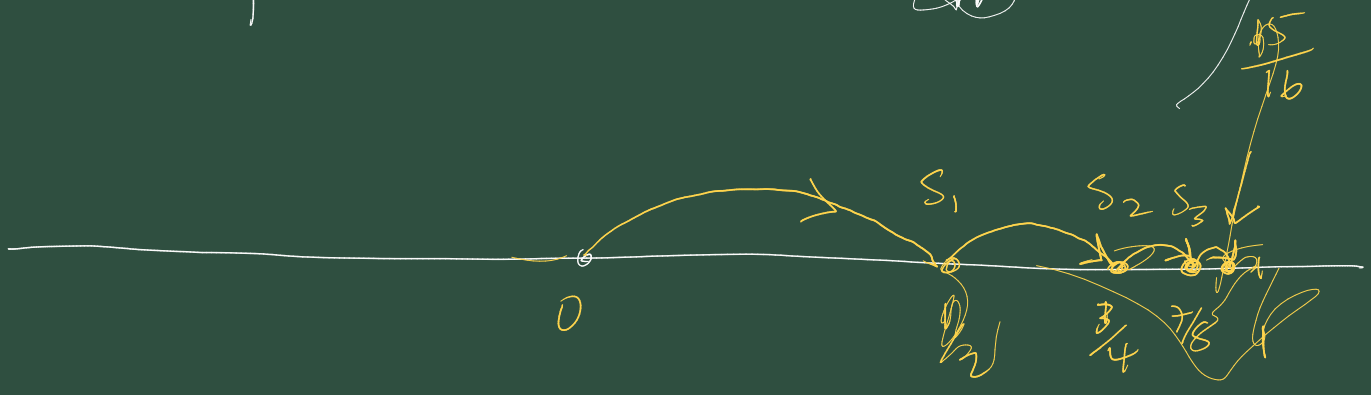
in fact $$\sum_{m=1}^{k} m = \frac{k}{2}(k+1) = \frac{k^2}{2} + \frac{k}{2}$$

and $\frac{k^2}{2} + \frac{k}{2}$ clearly diverges (increases without upper bound) as $k \to \infty$

Ex 2.2.3 A convergent series

Consider $$\sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

Let's plot the partial sums $\sum_{m=1}^{k} \left(\frac{1}{2}\right)^m$



This process (sequence of partial sums S_k) is converging to 1.

So it seems $$\sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m = 1.$$

(we can prove this formally using geometric series test).

From these two examples

- If the terms are too big, series diverges
- If the terms are converging to 0 small, maybe the series can converge

↳ Maybe this could be a theorem

Ex 2.2.4. The Harmonic Series

Consider $\sum_{m=1}^{\infty} \frac{1}{m} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$

A grouping argument can show that this diverges.

$$\sum_{m=1}^{\infty} \frac{1}{m} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)$$

$$+ \left(\frac{1}{9} + \dots + \frac{1}{16}\right) + \left(\frac{1}{17} + \dots + \frac{1}{32}\right)$$

$$+ \dots\dots \quad \text{(doubling the group size each time)}$$

$$> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right)$$

$$+\underbrace{\left(\frac{1}{16}+\dots+\frac{1}{16}\right)}_{8\text{ terms}}+\underbrace{\left(\frac{1}{32}+\dots+\frac{1}{32}\right)}_{16\text{ terms}}$$

$$+\dots$$

$$=1+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\dots$$

which is clearly divergent, increasing without any upper bound.

and so $\sum_{m=1}^{\infty}\frac{1}{m}$ diverges also.

(remember $\sum_{m=1}^{\infty}\frac{1}{2^m}$ converged).

So terms $\to 0$ is not enough to make the series converge.

Main steps $\sum a_m = A$, $\sum_{m=1}^{\infty} b_m = B$

$$\Rightarrow \sum(\alpha a_m+\beta b_m) = \alpha A+\beta B.$$

$$\alpha A+\beta B = \alpha \lim_{k\to\infty}\left(\sum_{m=1}^{k} a_m\right)+\beta \lim_{k\to\infty}\left(\sum_{m=1}^{k} b_m\right)$$

$$= \lim_{k\to\infty} \left(\alpha \left(\sum_{m=1}^{k} a_m \right) + \beta \left(\sum_{m=1}^{k} b_m \right) \right)$$

, by algebra of limits for sequences.

$$= \lim_{k\to\infty} \left(\sum_{m=1}^{k} \left(\alpha a_m + \beta b_m \right) \right)$$, rearranging the sum

$$= \sum_{m=1}^{\infty} \left(\alpha a_m + \beta b_m \right)$$; by definition of $\sum_{1}^{\infty}$.

If $\sum_{m=1}^{\infty} a_m = A$, $\sum_{m=1}^{\infty} b_m = B$

then what is

$$\sum_{m=1}^{\infty} a_m b_m = ?$$

$$AB = \left(\sum_{m=1}^{\infty} a_m \right) \left(\sum_{m=1}^{\infty} b_m \right)$$

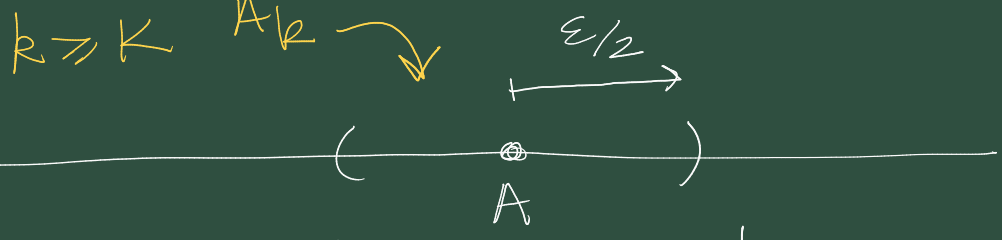
$$= \left(a_1 + a_2 + \dots \right) \left(b_1 + b_2 + \dots \right)$$

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General Term Test

Proof Assume $\sum_{m=1}^{\infty} a_m$ converges, say $\sum_{m=1}^{\infty} a_m = A$. Let $A_k = \sum_{m=1}^{k} a_m$, and $A_k \to A$ as $k \to \infty$.

So given any $\varepsilon > 0$, there exist a point K such that for all $k \geq K$ $|A_k - A| < \frac{\varepsilon}{2}$



$\Rightarrow$ for all $k \geq K$ $|A_k - A_{k+1}| < \varepsilon$

$\Rightarrow$ for all $k \geq K$ $|-a_{k+1}| < \varepsilon$

$\Rightarrow$ for all $k \geq K$ $|a_{k+1} - 0| < \varepsilon$

$\Rightarrow \lim_{n \to \infty} a_n = 0$, by the definition of a sequence convergence.

looking again at our first example

$\sum_{n=1}^{\infty} n$, diverges by the General Term Test.

We completely understand geometric series and can say for definite whether one converges / diverges and what the limit is.

Geometric Series Test

Consider $\sum_{n=1}^{\infty} ar^{n-1}$, for $a \neq 0$, converges if and only if $|r| < 1$, in which case

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}.$$

Proof If $a = 0$

$$\sum_{n=1}^{\infty} ar^{n-1} = 0 + 0 + 0 + 0 + \cdots = 0$$

Assume $a \neq 0$., If $r = 1$

then $\sum_{n=1}^{\infty} ar^{n-1} = a + a + a + \cdots$ clearly divergent

Otherwise, $a \neq 0$ and $r \neq 1$

$$S_k = \sum_{n=1}^{k} ar^{n-1} = a + ar + ar^2 + \cdots + ar^{k-1}$$

$$rS_k = \sum_{n=1}^{k} ar^n = ar + ar^2 + ar^3 + \cdots + ar^{k-1} + ar^k$$

$$(1-r)S_k = a - ar^k \quad \text{, by cancellation.}$$

$$\Rightarrow S_k = \frac{a(1-r^k)}{1-r}$$

$$\Rightarrow \lim_{k\to\infty} S_k = \frac{a\left(1 - \lim_{k\to\infty} r^k\right)}{1-r}$$

if $|r|<1$ then $r^k \to 0$ as $k\to\infty$

if $|r|>1$ then r^k diverges as $k\to\infty$

if $r=-1$ then r^k diverges as $k\to\infty$

So $\sum_{n=1}^{\infty} ar^{n-1}$ $(a\neq 0, r\neq 1)$ converges if and only if $|r|<1$

in which case

$$\sum_{n=1}^{\infty} ar^{n-1} = \lim_{k\to\infty} S_k = \frac{a}{1-r}.$$

Examples Consider again the second example today.

$$\sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m$$

$$= \sum_{m=1}^{\infty} \frac{1}{2}\left(\frac{1}{2}\right)^{m-1}$$ is a geometric series with $a = \frac{1}{2}$, $r = \frac{1}{2}$

so this converges by the test

$$= \frac{1/2}{1 - 1/2}$$

$$= 1.$$

Consider $\sum_{n=1}^{\infty} x_n$ where.

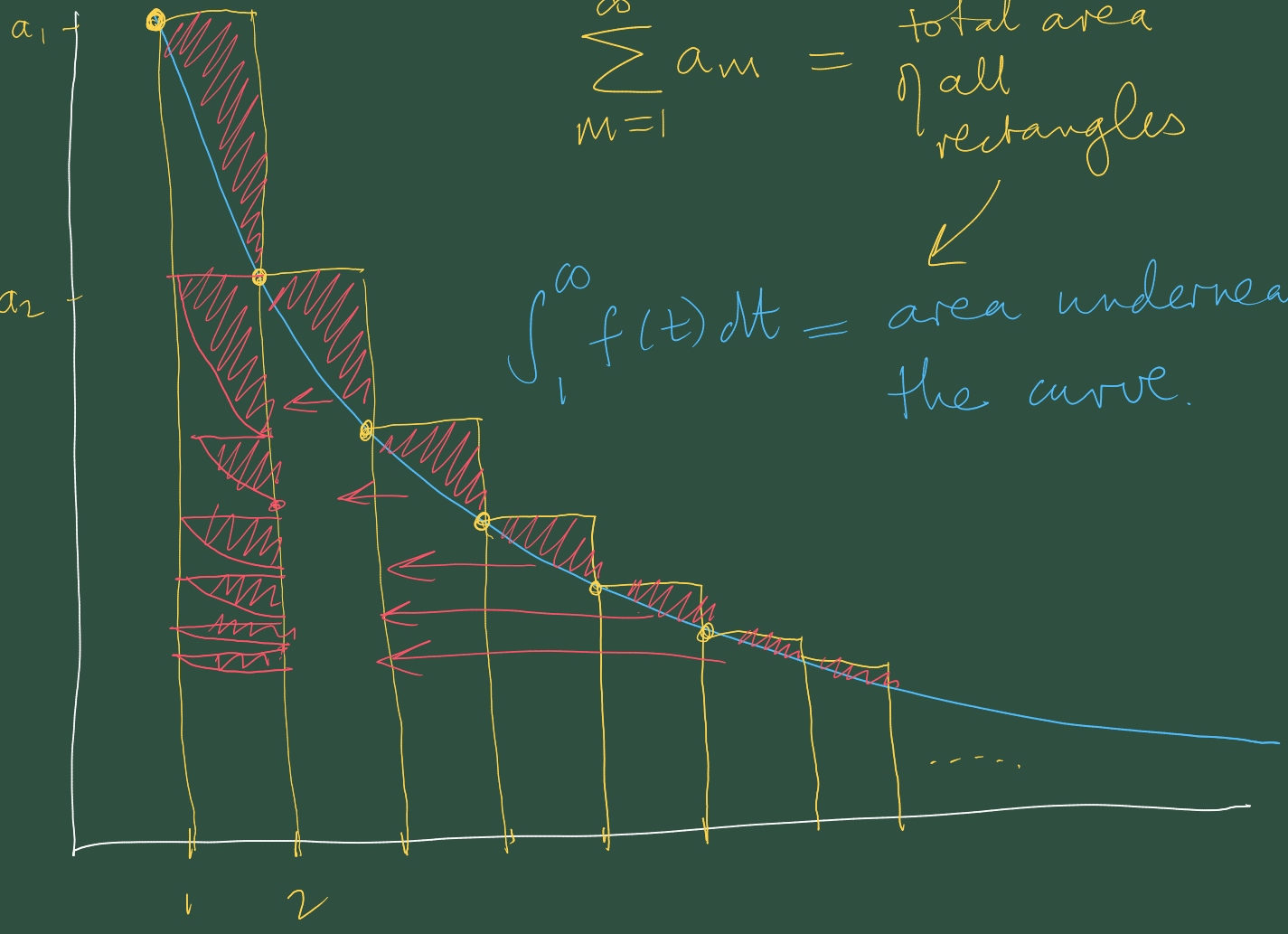
$$x_n = \frac{2}{3}\left(\frac{1}{2}\right)^n + \frac{1}{4}\left(\frac{2}{3}\right)^{n+1}$$

$$= \frac{1}{3}\left(\frac{1}{2}\right)^{n-1} + \frac{1}{9}\left(\frac{2}{3}\right)^{n-1}$$

So x_n is a sum of two convergent geometric series. $\sum_{n=1}^{\infty} \frac{1}{3}\left(\frac{1}{2}\right)^{n-1} = \frac{1/3}{1 - 1/2} = \frac{2}{3}$

$$\sum_{n=1}^{\infty} \frac{1}{9}\left(\frac{2}{3}\right)^{n-1} = \frac{1/9}{1 - 2/3} = 1/3$$

So by algebra/linear combinations result for series. $\sum_{n=1}^{\infty} x_n = \frac{2}{3} + \frac{1}{3} = 1.$



$$\sum_{m=1}^{\infty} a_m = \text{total area of all rectangles}$$

$$\int_1^{\infty} f(t)\,dt = \text{area underneath the curve.}$$

$$\sum_{m=1}^{\infty} a_m \approx \int_1^{\infty} f(t)\,dt.$$

$$= \int_1^{\infty} f(t)\,dt + \underbrace{\text{total red area.}}_{\text{finite.} < a_1}$$

$$\sum_{m=1}^{\infty} a_m = \int_1^{\infty} f(t)\,dt + \gamma, \qquad \gamma \in \mathbb{R}.$$

This applies to our harmonic series

$$\sum_{m=1}^{\infty} \frac{1}{m}$$

and the continuous function

$$f(t) = \frac{1}{t}$$

and $\int_1^\infty \frac{1}{t}\,dt = \lim_{k\to\infty} \int_1^k \frac{1}{t}\,dt$

$$= \lim_{k\to\infty} \left[\ln(t)\right]_1^k$$

$$= \lim_{k\to\infty} \ln(k) \text{ , which is divergent}$$

$$\Rightarrow \sum_{m=1}^{\infty} \frac{1}{m} \text{ diverges also, by the integral test.}$$